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TECHNICAL UNIVERSITY OF CLUJ-NAPOCA ROMANIA

Project at CAD Techniques

Weight Control System for Storage Shelf

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I. Context

1. Requirements

Specifications:

Design an electronic monitoring and control system for the weight supported by a storage shelf, using a resistive sensor whose resistance varies linearly with the detected mass. The circuit must convert the sensor resistance into a linear electrical voltage mapped within the range of 0 V to $V_{CC} - 2V$, and use a comparator stage to detect whether the weight is within or outside the safety interval $[W_L, W_H]$. The system, powered by the supply voltage V_{CC}, shall provide optical status indication through an LED with a current-limiting resistor, activating it to indicate underload, normal operation, or overload conditions according to the specified requirements.

				Led Colour for each condition		
Measurable weight range [kg]	The weight of the shelf [kg]	Sensor resistance $[k\Omega]$	V_{CC} [V]	Underload condition	Normal	Overload condition
7 - 250	40 - 200	1 - 21	10	Red	Orange	Yellow

Given data:

Sensor measurable range [kg]: $W_{min} = 7\ kg$, $W_{max} = 250\ kg$

Safe operating range [kg]: $W_L = 40\ kg$, $W_H = 200\ kg$

Sensor resistance range [kg]: $R_{min} = 1\ k\Omega$, $R_{max} = 21\ k\Omega$

Supply voltage [V]: $V_{cc} = 10\ V$

Led signaling colors:

- Red: Underload condition (below 40 kg)
- Orange: Normal condition (between 40 kg and 200 kg)
- Yellow: Overload condition (above 200 kg)

II. About the Schematic

2. Schematic Description

2.1. Theoretical Support

Weight Sensor

The weight sensor is like a resistor that changes how much it resists electricity when something is put on the shelf. The more weight is on the shelf the more the sensor resists electricity. The weight sensor is represented in Figure 1.

We need to turn this change in resistance into a signal that we can measure. So we use a setup with two resistors, one that changes and one that stays the same. The changing resistor is the weight sensor and the fixed resistor is like a helper. They are connected together in a line with a power source. This setup gives us a signal that changes when the sensor resistance changes. The signal is like a message that tells us how much weight is on the shelf. The relationship between the weight W and the sensor resistance R is described by a linear function, according to relation (1):

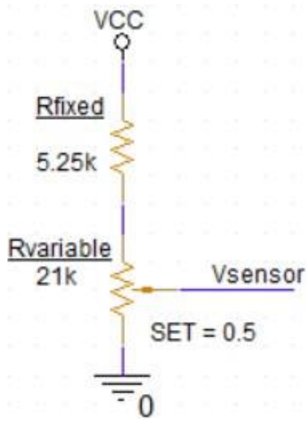
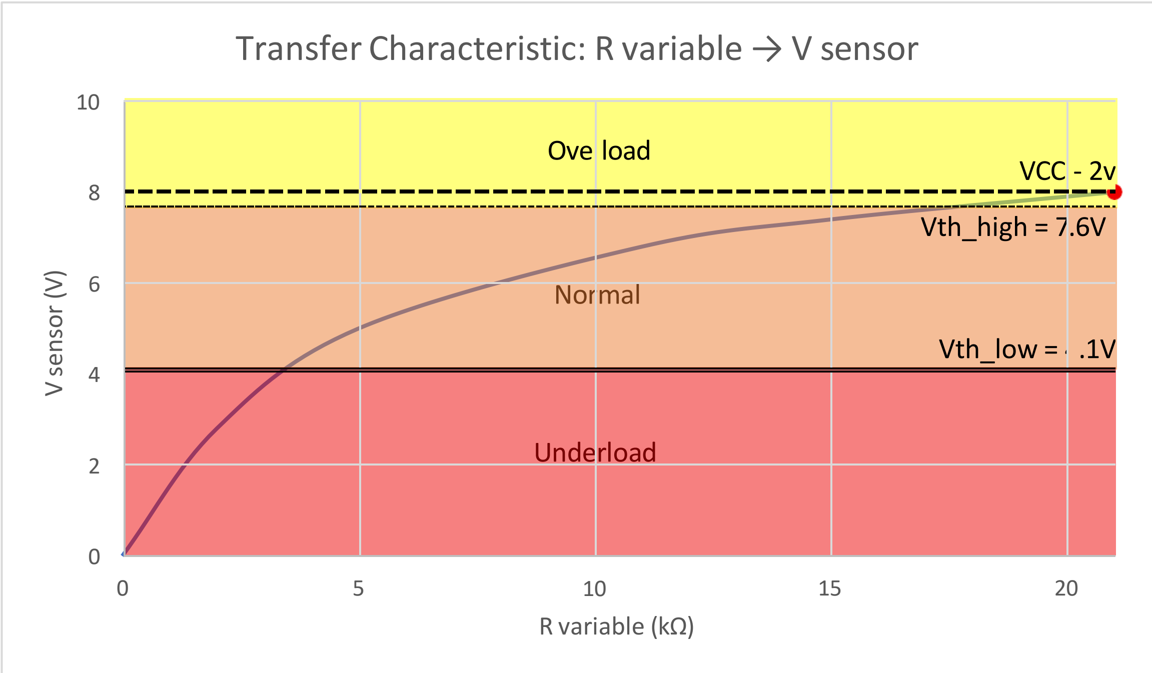


Figure 1. The Weight Sensor

$$R(W) = R_{min} + m_R \cdot (W - W_{min}) = R_{min} + \frac{R_{max} - R_{min}}{W_{max} - W_{min}} \cdot (W - W_{min}) \qquad (1)$$

R_{min} is the resistance when we have the weight. R_{max} is the resistance when we have the weight. This means that the way the resistance changes is linear. This is good because it makes it wasy to figure out the voltage. The physical changes and the electrical signal are directly related to each other, that is why we use R_{min} and R_{max}, to understand this relationship.



Resistance to Voltage Conversion

In this section we change the sensor resistance caused by the weight into a voltage that the circuit can process. This is done using a voltage divider made of a fixed resistor (R_{fixed}) and the sensor resistance ($R_{variable}$), as shown in Figure 2. As the sensor resistance ($R_{variable}$) changes the output voltage (V_{sensor}) also changes according to relation (2):

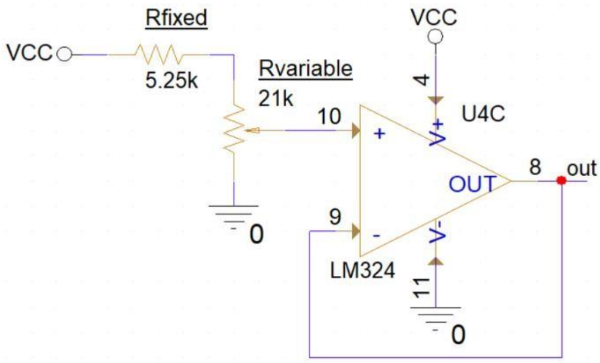


Figure 2. Resistance to Voltage Conversion

$$V_{sensor} = V_{cc} \cdot \frac{R_{variable}}{R_{variable} + R_{fixed}} \quad (2)$$

$$V_{sensor}(at\ R_{min} \approx 5.25k\Omega) = 10 \cdot \frac{5.25}{5.25 + 5.25} = 5V \quad (3)$$

$$V_{sensor}(at\ R_{max} \approx 21k\Omega) = 10 \cdot \frac{21}{21 + 5.25} = 8V \quad (4)$$

The design need to map the full sensor resistance range to a voltage range of 0 to $V_{cc} - 2V$. The upper limit of $V_{cc} - 2V$ is set by the LM324 amplifiers characteristics. In single-supply configuration it cannot drive the output close to the supply voltage about 1.5V - 2V. From the condition that R_{max} should give $V_{cc} - 2V$, we calculate R_{fixed} using relation (5):

$$R_{fixed} = R_{variable} \cdot \frac{V_{cc} - V_{sensor}}{V_{sensor}} \tag{5}$$

$$R_{fixed} = 21k\Omega \cdot \frac{10V - 8V}{8V} = 21 \cdot \frac{2}{8} = 5.25k\Omega \tag{6}$$

Impedance Matching (The Voltage Follower)

A voltage follower (also called an unity-gain amplifier, a buffer amplifier and an isolation amplifier) is a op-amp circuit which has a voltage gain of 1, as shown in Figure 3.

This mean the op-amp does not make the signal any stronger. The reason it is called a voltage follower is because the output voltage directly follow the input voltage, meaning the output voltage is the same as the input voltage.

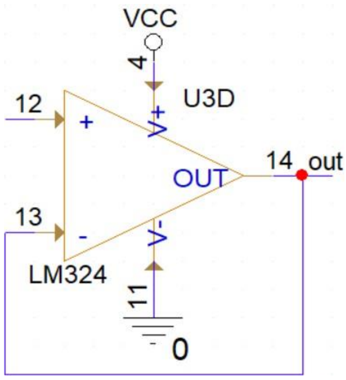


Figure 3. The voltage follower

The gain of the voltage follower:

$$V_{out} = V_{in} \rightarrow A_v = 1 \tag{7}$$

The Comparators

A window detector (window comparator) uses two LM324 operational amplifiers working without feedback as voltage comparators, as presented in Figure 4. The circuit defines a safe operation range using two tresholds, V_L, V_H created using resistors R_6, R_7, R_8. The sensor voltage $V_{set}$is compared with these tresholds to determine three conditions:

- Underload Zone: When $V_{set} < V_{th\_low}$, the lower comparator activates;
- In Range (Normal) Zone: When $V_{th\_low} \leq V_{set} \leq V_{th\_high}$, both comparators output 0V, meaning the voltage is sitting in the safe „window”;
- Overload Zone: When $V_{set} > V_{th\_high}$, the upper comparator activates.
- The relationships for the voltage tresholds are given by equation (8) and (10):

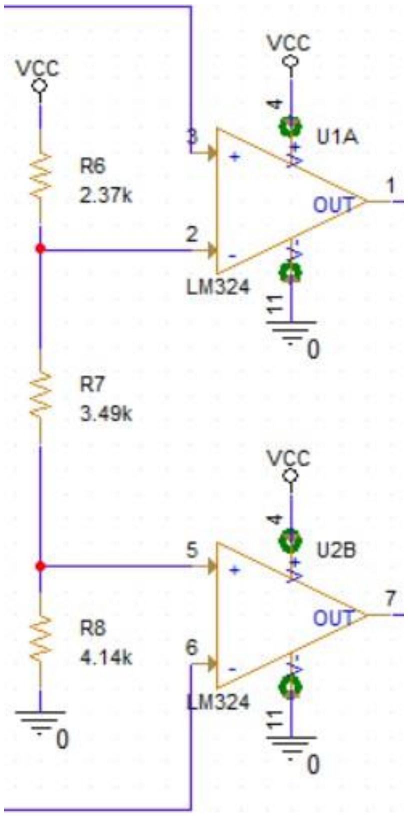


Figure 4. The comparators

$$V_{th\_high} = V_{cc} \cdot \frac{R_7 + R_8}{R_6 + R_7 + R_8} \tag{8}$$

$$V_{th\_high} = 10 \cdot \frac{3.49 + 4.14}{2.37 + 3.49 + 4.14} = 7.64V \tag{9}$$

$$V_{th\_low} = V_{cc} \cdot \frac{R_8}{R_6 + R_7 + R_8} \tag{10}$$

$$V_{th\_low} = 10 \cdot \frac{4.14}{2.37 + 3.49 + 4.14} = 4.14V \tag{11}$$

Calculations to find the appropriate values for $R_6 + R_7 + R_8$:

$$I = \frac{V_{cc}}{R_{total}} = \frac{10V}{10k\Omega} = 1mA \tag{12}$$

$$R_6 = \frac{V_{cc} - V_H}{I} = \frac{10V - 7.63V}{1mA} = \frac{2.37V}{1mA} = 2.37k\Omega \tag{13}$$

$$R_7 = \frac{V_H - V_L}{I} = \frac{7.63V - 4.14V}{1mA} = 3.49k\Omega \tag{14}$$

$$R_8 = \frac{V_L}{I} = \frac{4.14V}{1mA} = \frac{4.14V}{1mA} = 4.14k\Omega \tag{15}$$

$$R_6 + R_7 + R_8 = 2.37 + 3.49 + 4.14 = 9.99k\Omega \approx 10k\Omega \tag{16}$$

OR Logic and Transistor Divider

Diodes $D_4$and D_5(1N4148) implemented a wired OR function by combining the outputs of U10A and U2B that drives the Q_1 (Q2N2222) transistor.The diodes prevent short-circuiting between comparato outputs while allowing either overload and underload signals to control the circuit. When wither output is HIGH, the base current flows through R_5 (10kΩ), turning

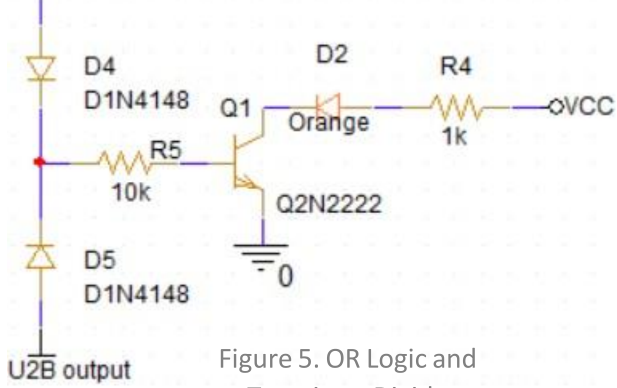
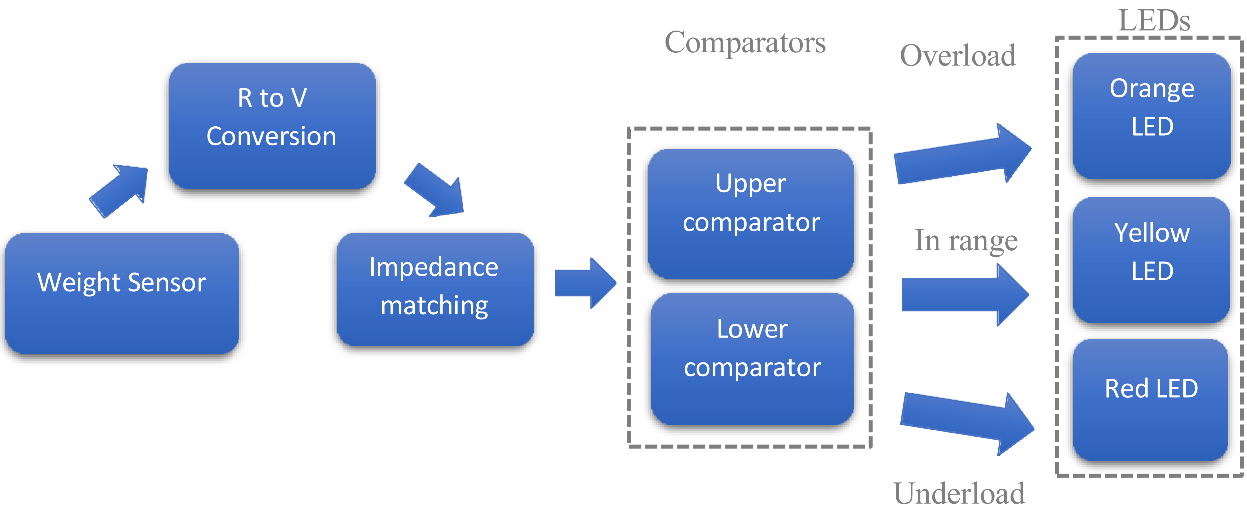


Figure 5. OR Logic and Transistor Divider

Q_1on. The transistor acts as a switch, grounding the orange LED (D_2), which is powered from V_{cc}through R_4(1kΩ). The deschibed circuit is implemented in Figure 5.

2.2. Block Diagram



2.2.1. Block 1 - Weight Sensor

The weight sensor represents the physical input of the system. It is modeled by a variable resistor $R_{variable}$whose resistance varies linearly with the mass applied to the sheld, accoeding to relationship (1).The nominal value in the simulation is 21 kΩ, corresponding to the maximum weight in the ranfe. The SET parameter in the schematic allows adjusting the resistance value to simulate different load conditions.

2.2.2. Block 2 - R to V Conversion

The resistance-voltage conversion is performed by a resistive divider consisting of R_{fixed} (5.25 kΩ) connected between V_{cc} and the sensor node, and $R_{variable}$connected between the same node and ground. The sensor voltage V_{sensor} is The output voltage is taken from the common node, as shown in equation (2). The value of R_{fixed} was calculated so that R_{max}corresponds to the voltage $V_{cc} - 2V = 8V$, and $R_{min}(\approx 0\Omega\, in\, empty\, shelf\, condition)$ corresponds to 0V.

2.2.3. Block 3 - Impedance matching

U3C is set uo as a unity-gain voltage follower (voltage repeater) according to equation (4). The non-inverting input (pin 10) receives V_{sensor}, and the output (pin 8) is connected directly to the inverting input (pin 9). The output V_{set} is the same as V_{sensor} but it comes from the operational amplifiers output stage, with extremely low output impedance. This block is essential to prevent errors caused by the load on the resistive divider.

2.2.4. Block 4 – Comparators

Both comparators U1A and U2B receive V_{set} on the inverting input (-). The non-invertiong (+) input receive the treshold voltages generated by the ladder divider $R_6 - R_7 - R_8$, according to equations (7) and (8). The three output states are shown in Table 1:

Condition	**U1A output**	**U2B output**	**Red LED**	**Orange LED**	**Yellow LED**
$V_{sensor} < V_{th\_low}$	LOW	LOW	OFF	OFF	ON
$V_{th\_low} \leq V_{sensor} \leq V_{th\_high}$	LOW	HIGH	OFF	ON	OFF
$V_{sensor} > V_{th\_high}$	HIGH	HIGH	ON	OFF	OFF

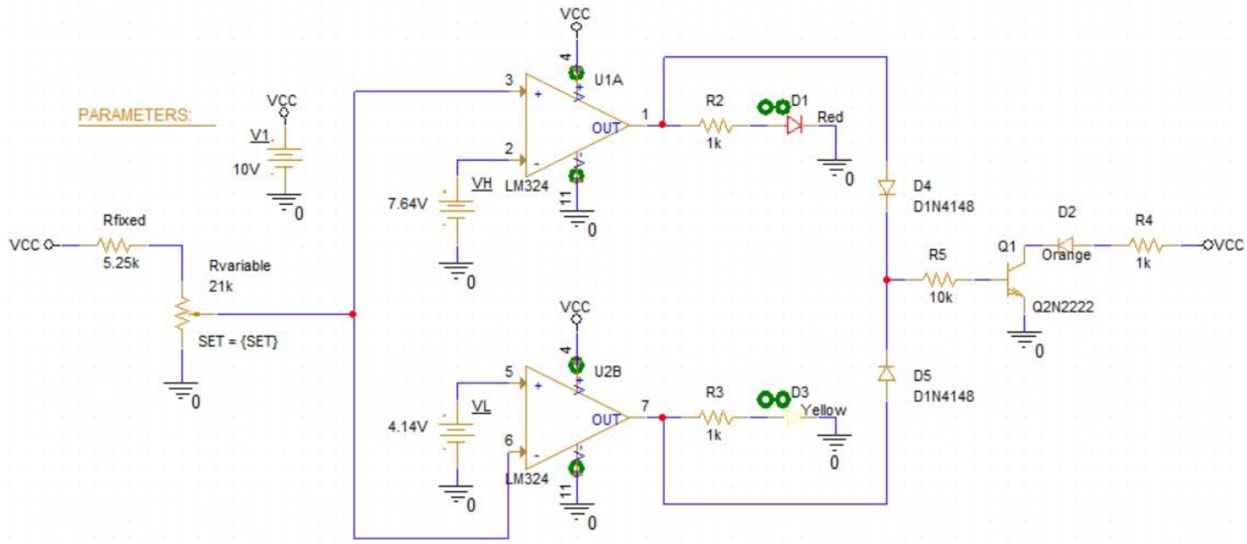
Table 1. Logic States of the Window Detector

2.2.5. Block 5 - LEDs

The output stage changes the logic levels of the comparators into visual indications. The red LED D_1 and resistor $R_2(2k\Omega)$ are controlled by the output of U1A. The yellow LED D_3 and $R_3(1k\Omega)$ are controlled by the output U2B. The orange LED D_2 need a transsitor stage Q_1(Q1N2222) because it only turns on when both comparator outputs are LOW. The diodes D_4 and D_5 which are 1N4148 help with the wired OR function. If either comparator output is HIGH, the base of the transistor is biased and Q_1 conducts, which turn off D_2.

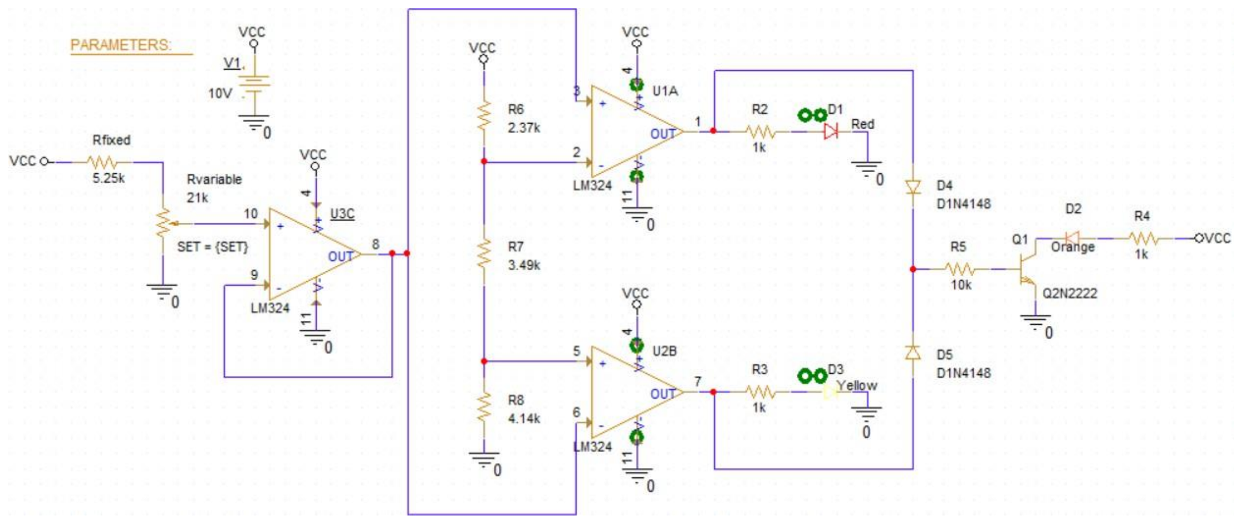
3. Electrical Schematic & Simulation Results

3.1. Old Electrical Schematic



The original circuit was made as a window comparator, it used two ideal voltage sources to set the fixed reference thresholds at 7.46V and 4.14V. The input voltage is adjusted with a potentiometer connected directly to the comparator inputs. This approach works in theory but it is not practical for real-world manufacturing and the reason is that using seperate batteries is expensive and not efficient, and also the potentiometer is not isolated so leakage currents or noise from the comparators can affect the voltage, that leads to readings and unstable behavior.

3.2. Updated Electrical Schematic



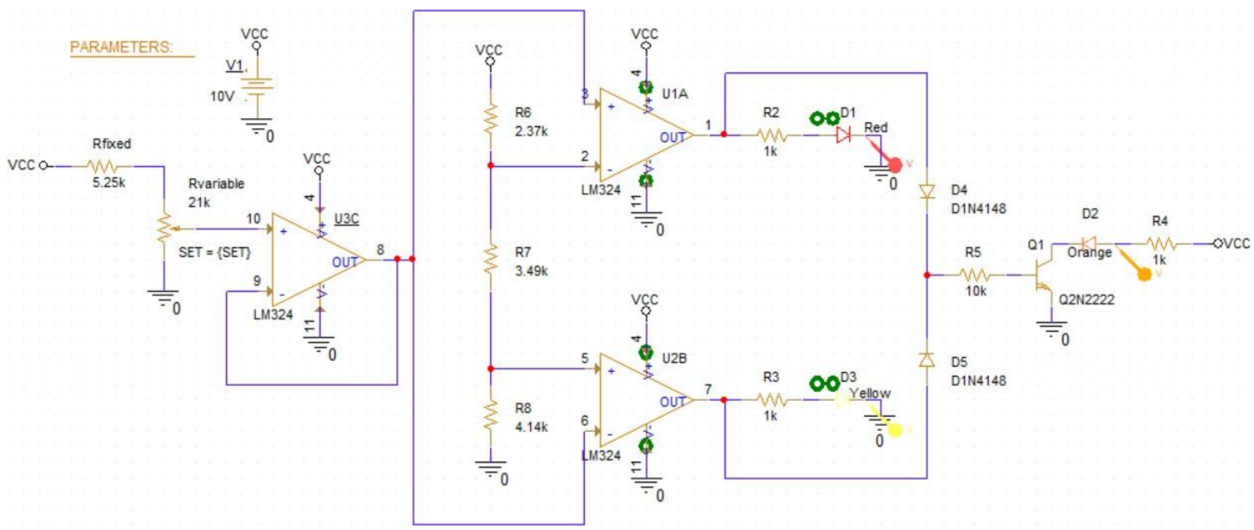
The new circuit fixed some problems that were happening by introducing two major upgrades for stability and efficiency. First, an operational amplifier configured as a voltage follower was added right after the potentiometer to act as a buffer. This buffer isolates the tuning stage from the rest of the circuit, ensuring the control voltage remains perfectly stable regardless of downstream behavior. Second, the separate ideal batteries were replaced by a single resistor chain powered directly from the main 10V supply. This not only simplifies the physical build but also creates a ratiometric system where any fluctuations in the main power source will cause both the reference thresholds and the tuning voltage to shift together proportionally, preventing any false triggering.

How it works?

The 10Vsupply voltage (V_{cc}) powers all LM324 units and LEDs. The resistive sensor and fixed resistor form a voltage divider, generating V_{sensor}, which is buffered by U3C and applied to both comparators. Reference voltages V_{th_low} and V_{th_high} are generated by the $R_6 - R_7 - R_8$ divider and connected to the comparator inputs. The comparator outputs drive the red and yellow LEDs through current-limiting resistors and also control transistor Q1 through diodes D_4, D_5, and resistor R_5.

3.3. Simulation Results

The ciruit:



DC Sweep Analysis

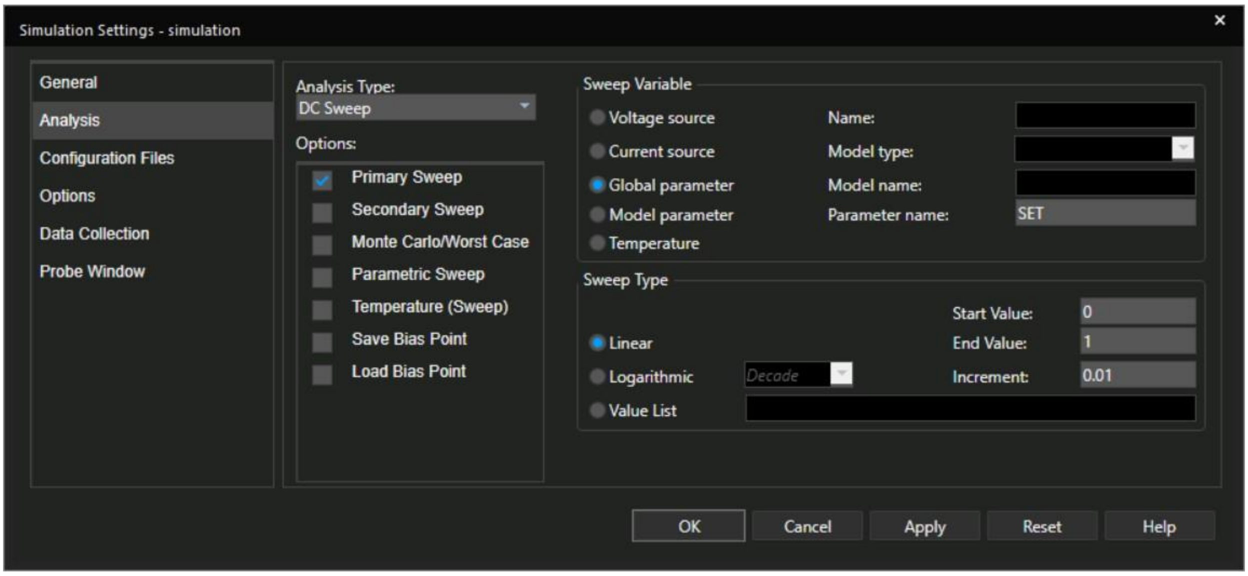


Figure 6. Simulation Profile

To simulate the circuit, we put Global Parameter named exactly SET, with variation type Linear, starting value 0, final value 1, and increment of 0.01, to completely sweep the potentiometer slider and observe the switching of pin 1 and pin 7 outputs.

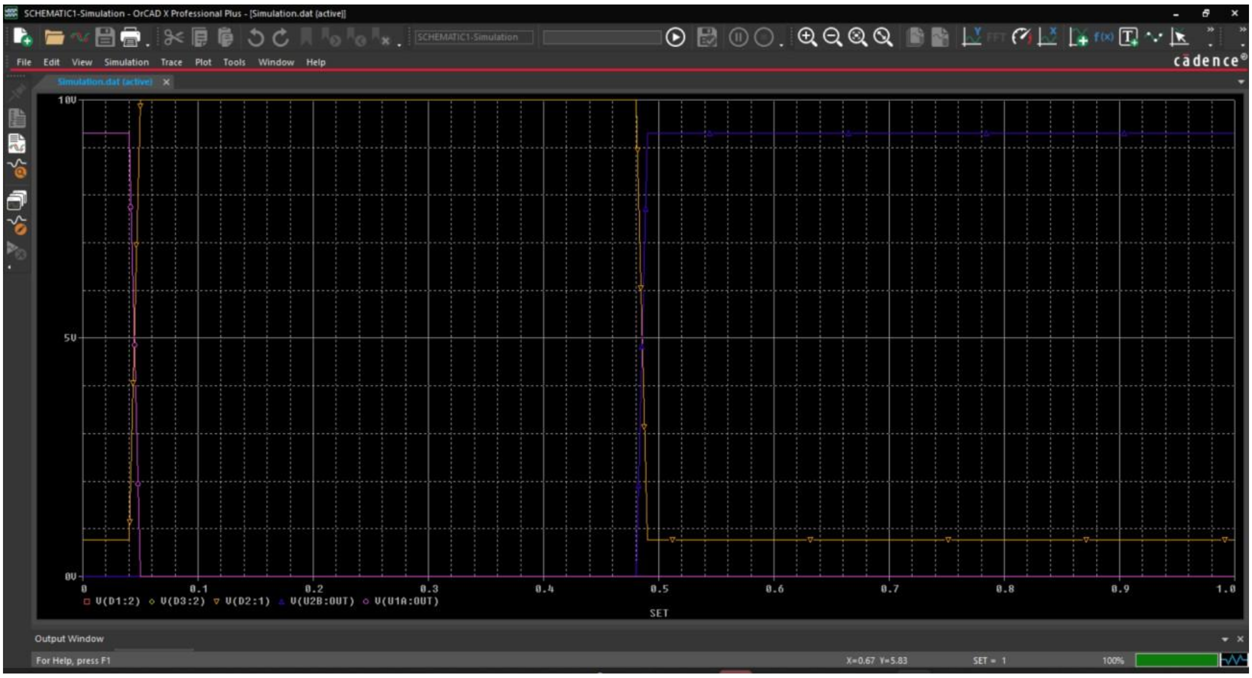


Figure 7. Simulation Result

The X-axis (SET) represents the simulated weight (increasing the sensor's resistance), and the Y-axis represents the output voltage driving the LEDs.

A. Underload Condition (Left side of the graph, SET < ~0.5)

The shelf weight is below the 40 kg minimum, so the sensor voltage drops under the lower threshold V_L. This makes comparator U1A output go HIGH (seen as the PINK trace V(U1A:OUT) rising to about 10 V). That HIGH signal drives the ORANGE trace V(D1:1), which settles around ~2 V, confirming the Red LED (D1) is ON and indicating an underload condition

B. Normal Operating Range (Middle of the graph, SET between ~0.5 and ~0.48)

The weight is within the safe 40–200 kg range, so the sensor voltage stays between the thresholds $V_L$and V_H. Both comparators remain LOW (seen as the PINK trace V(U1A:OUT) and BLUE trace V(U2B:OUT) near 0 V). As a result, the ORANGE trace V(D1:1) and YELLOW trace V(D2:1) also stay at 0 V, confirming both the Red and Yellow LEDs are OFF, indicating normal operation.

C. Overload Condition (Right side of the graph, SET > ~0.48)

The weight exceeds the 200 kg limit, so the sensor voltage rises above the upper threshold V_H. This makes comparator U2B output go HIGH (seen as the BLUE trace V(U2B:OUT) jumping to about 9–10 V). That HIGH signal drives the YELLOW trace V(D3:1) to around ~2 V, confirming the Yellow LED (D3) is ON and indicating an overload condition.

Underload Mode

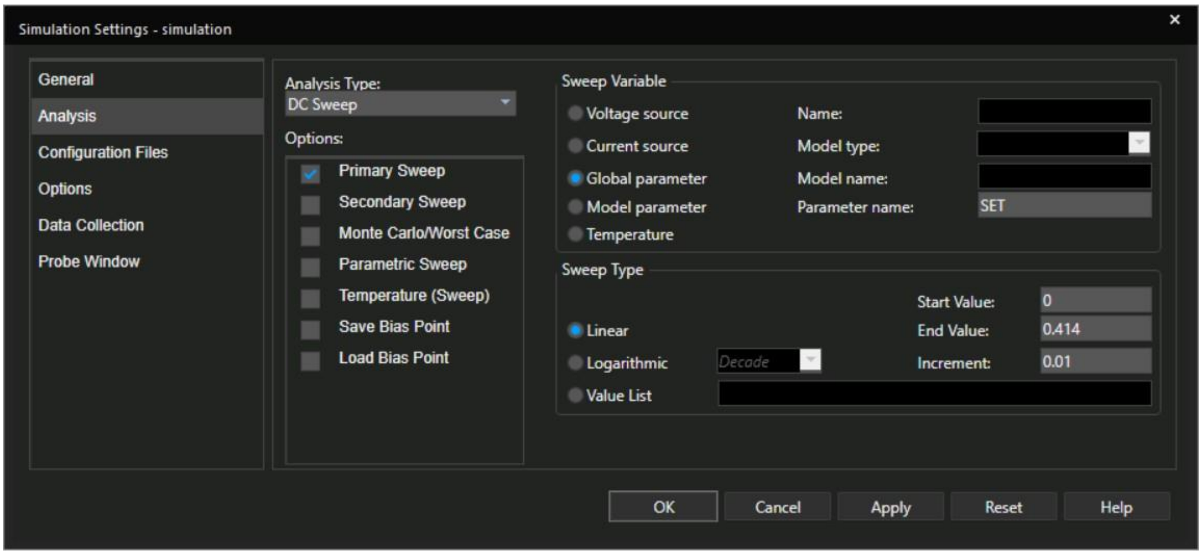
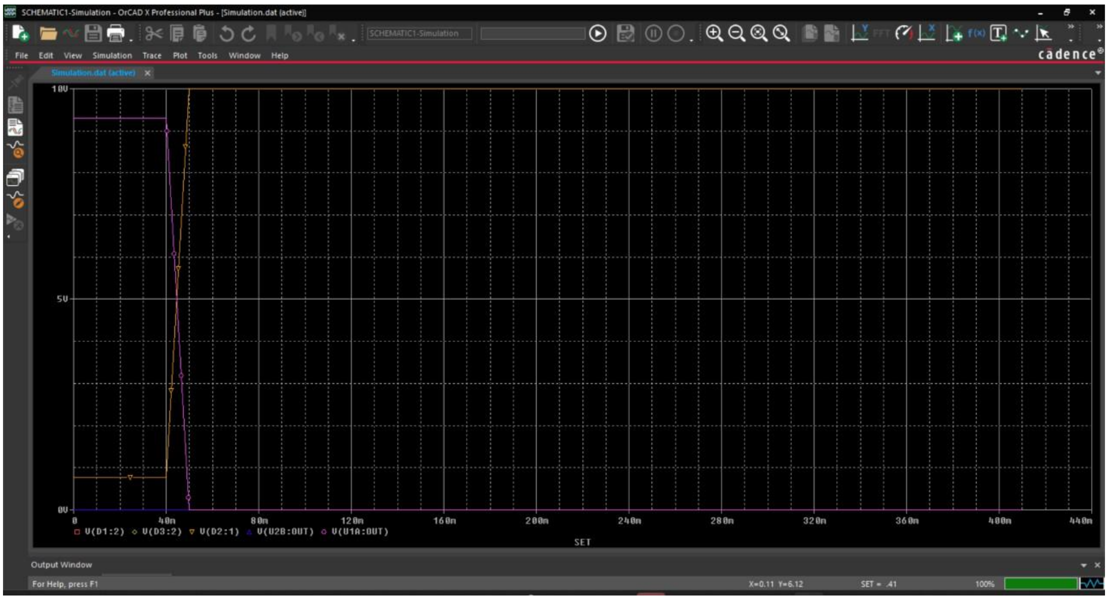


Figure 8. Simulation Profile



Normal

Figure 9. Simulation Result

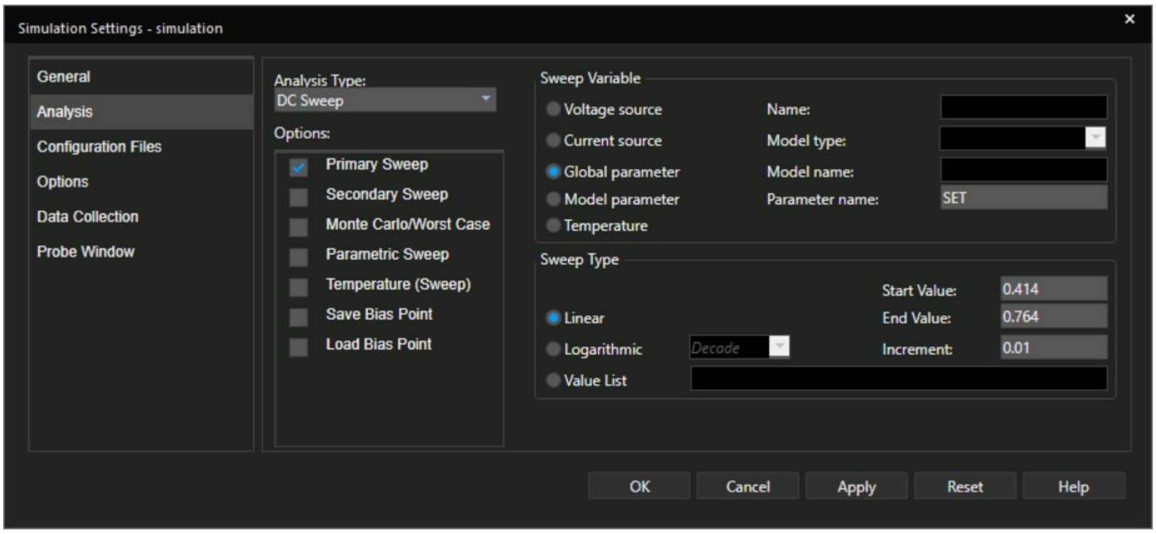


Figure 10. Simulation Profile

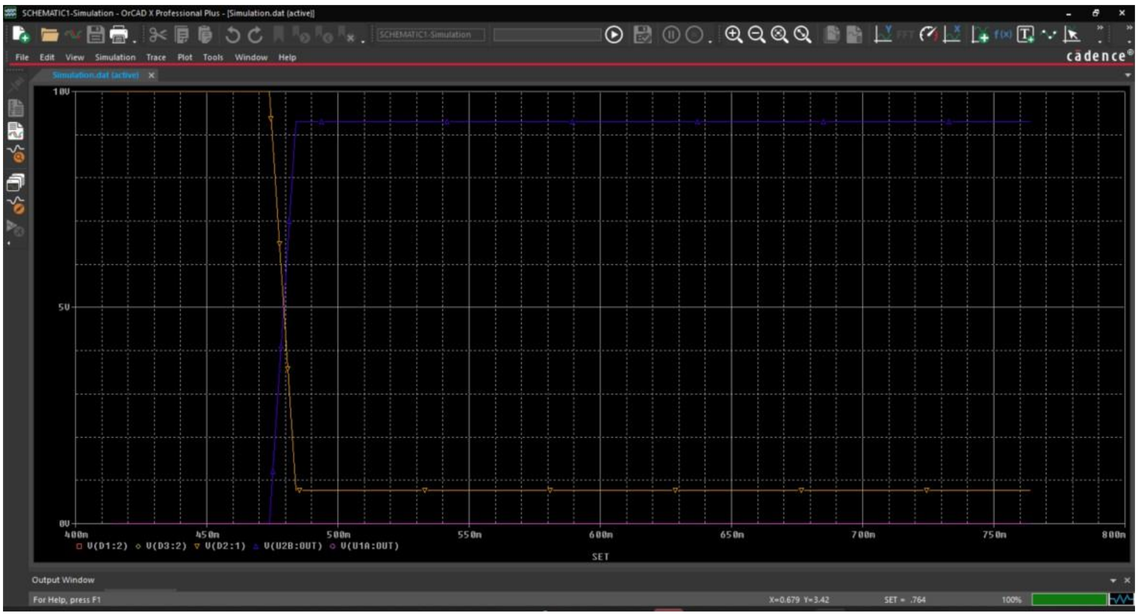


Figure 11. Simulation Result

Overload Mode

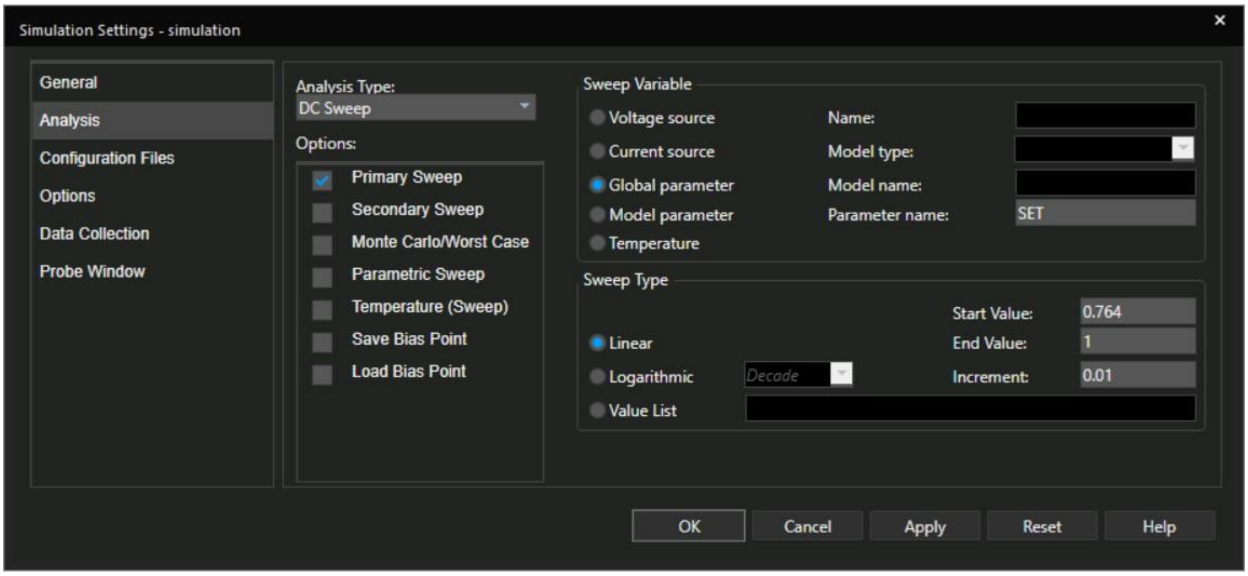


Figure 12. Simulation Profile

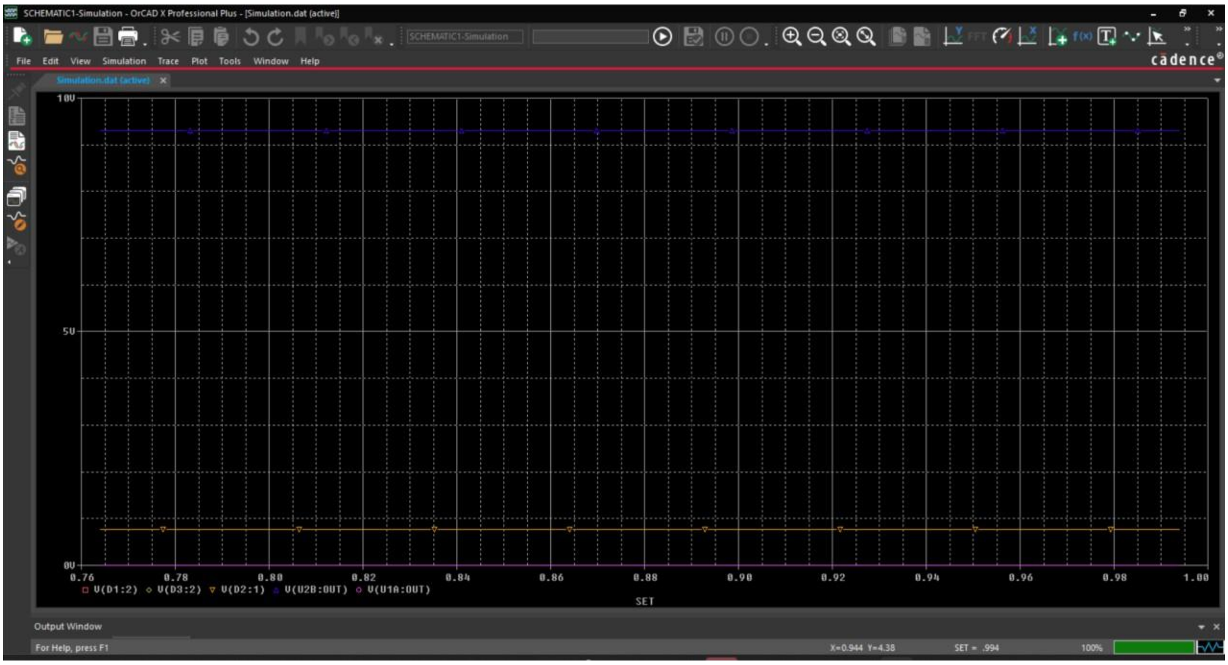


Figure 13. Simulation Result

3.4. Bill of Materials (BOM)

Table 2. Bill of materials(BOM)

Nr.	Ref.	Component	Value / Type	Qty	Unit cost	Function
1	V1	Voltage source	10 V DC	1	—	Main power supply
2	Rfixed	Resistor	5.25 kΩ, 1/4W 1%	1	~0.05 €	Voltage divider upper arm
3	Rvariable	Resistive sensor	0–21 kΩ, linear	1	~2.00 €	Weight transducer
4	R2	Resistor	1 kΩ, 1/4W 5%	1	~0.02 €	Red LED current limiter
5	R3	Resistor	1 kΩ, 1/4W 5%	1	~0.02 €	Yellow LED current limiter
6	R4	Resistor	1 kΩ, 1/4W 5%	1	~0.02 €	Orange LED current limiter
7	R5	Resistor	10 kΩ, 1/4W 5%	1	~0.02 €	Q1 base pull-up
8	R6	Resistor	2.37 kΩ, 1/4W 1%	1	~0.05 €	Threshold divider (upper)
9	R7	Resistor	3.49 kΩ, 1/4W 1%	1	~0.05 €	Threshold divider (middle)
10	R8	Resistor	4.14 kΩ, 1/4W 1%	1	~0.05 €	Threshold divider (lower)
11	U1A, U2B, U3C	Op-amp	LM324, DIP-14	1 pkg	~0.35 €	Comparators and buffer
12	Q1	NPN BJT	Q2N2222, TO-92	1	~0.08 €	Orange LED switch
13	D1	LED	Red 5 mm, Vf≈1.9 V	1	~0.05 €	Overload indicator
14	D2	LED	Orange 5 mm, Vf≈2.0 V	1	~0.05 €	Normal range indicator
15	D3	LED	Yellow 5 mm, Vf≈1.9 V	1	~0.05 €	Underload indicator
16	D4, D5	Diode	1N4148, DO-35	2	~0.04 €	Wired-OR isolation

The estimated total cost of components for a prototype is approximately €3.00–3.50, excluding PCB and connectors. The component with the highest relative cost is the resistive weight sensor (Rvariable ≈ €2.00), which in series production could be replaced by a Hall effect sensor or a FSR (Force Sensitive Resistor) transducer with superior characteristics.

3.5. Why these Components?

The components were selected to meet purely analog design constraints and the specific system parameters (Vcc = 10 V, sensor range 1–21 kΩ).

Potentiometer — 21 kΩ (Rvariable) Simulates the weight sensor, which the assignment defines as a linear resistance up to 21 kΩ at maximum load.

Rfixed — 5.25 kΩ Forms a voltage divider with the sensor. Chosen to cap Vout at ~8 V (Vcc − 2 V), preventing op-amp saturation.

LM324 (U3C, U1A, U2B**)** Chosen because it supports single-supply operation at 10 V, can sense inputs down to ground, and one quad package covers all three op-amp functions. Older op-amps like the LM741 require a dual ±supply, making them incompatible with this circuit.

R6, R7, R8 (2.37 kΩ / 3.49 kΩ / 4.14 kΩ) A single three-resistor chain generates both comparator reference voltages, precisely defining the upper and lower thresholds of the safe operating window.

R2, R3 — 1 kΩ Limit LED current to a safe ~6–8 mA, ensuring proper brightness without damaging the LEDs or op-amp outputs.

LEDs — red, yellow, orange Assigned per specification: red for underload, yellow for overload, orange for fault outside the normal range.

1N4148 (D4, D5) Isolate the two comparator outputs and form an analogue OR function to drive Q1. Preferred over the 1N4007, which is a mains rectifier — oversized and unnecessary here.

Q2N2222 + R5 10 kΩ (Q1) The transistor switches the orange LED since the op-amp cannot source enough current directly. The 10 kΩ base resistor ensures safe operation and full saturation.

Conclusion

In conclusion, we designed and validated a schematic which controls the weight on a storage shelf by utilizing a transducer system consisting of a resistive sensor and a 5.25 kΩ voltage divider, a voltage follower that keeps intact the mapped voltage domain, a window comparator that defines two thresholds (V_{th_high} ≈ 7.64 V and V_{th_low}≈ 4.14 V) to accurately detect three load states, and a fully analog, wired-OR output stage that provides mutually exclusive LED indication without digital circuits, while proposing future improvements like hysteresis, FSR sensors, numeric displays, and input protection.

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